Develop NLP Apps with Azure AI Language

Azure provides nlp through language services enabling developers to incorporate these advanced features into their data application

- ➔ **Case Study :**Azure AI Language Service:
  - My friend Tina got a job at a smart home device company
  - The company's products are rapidly gaining popularity in the marketplace
  - Customer service agents are overwhelmed with questions and support requests
  - Tina has been assigned to lead a team to develop an AI solution

Project Goals

Goals	VS.	Capabilities
Extract key details to identify the core issue and categorize the inquiry appropriately		Azure AI Language capabilities to extract key phrases, recognize named entities and PII
Help to understand customer issues, by summarizing long emails		Text Summarization features of Azure AI Language
Address issues across a multi-language customer base		Cross-lingual capabilities of Azure AI Language

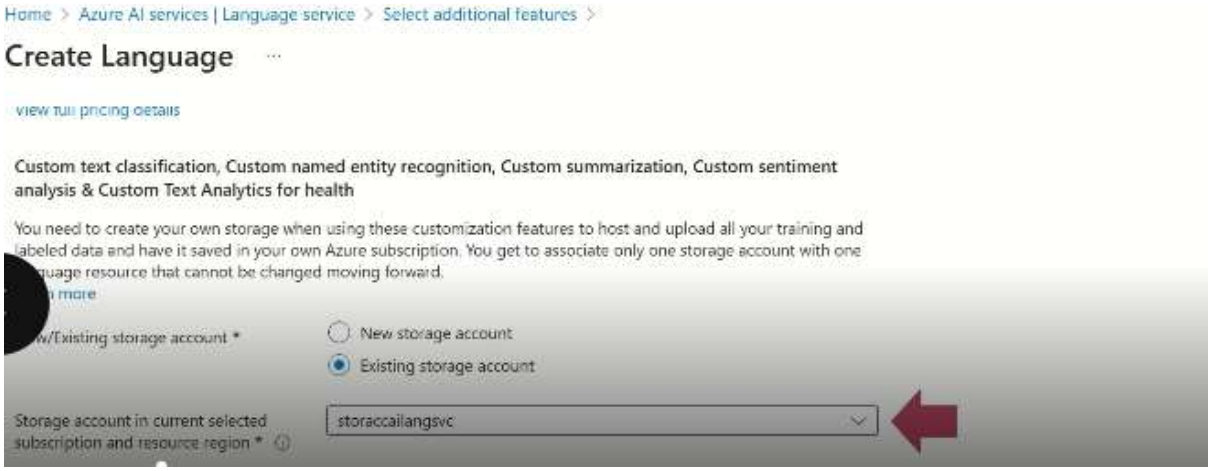
DEMO

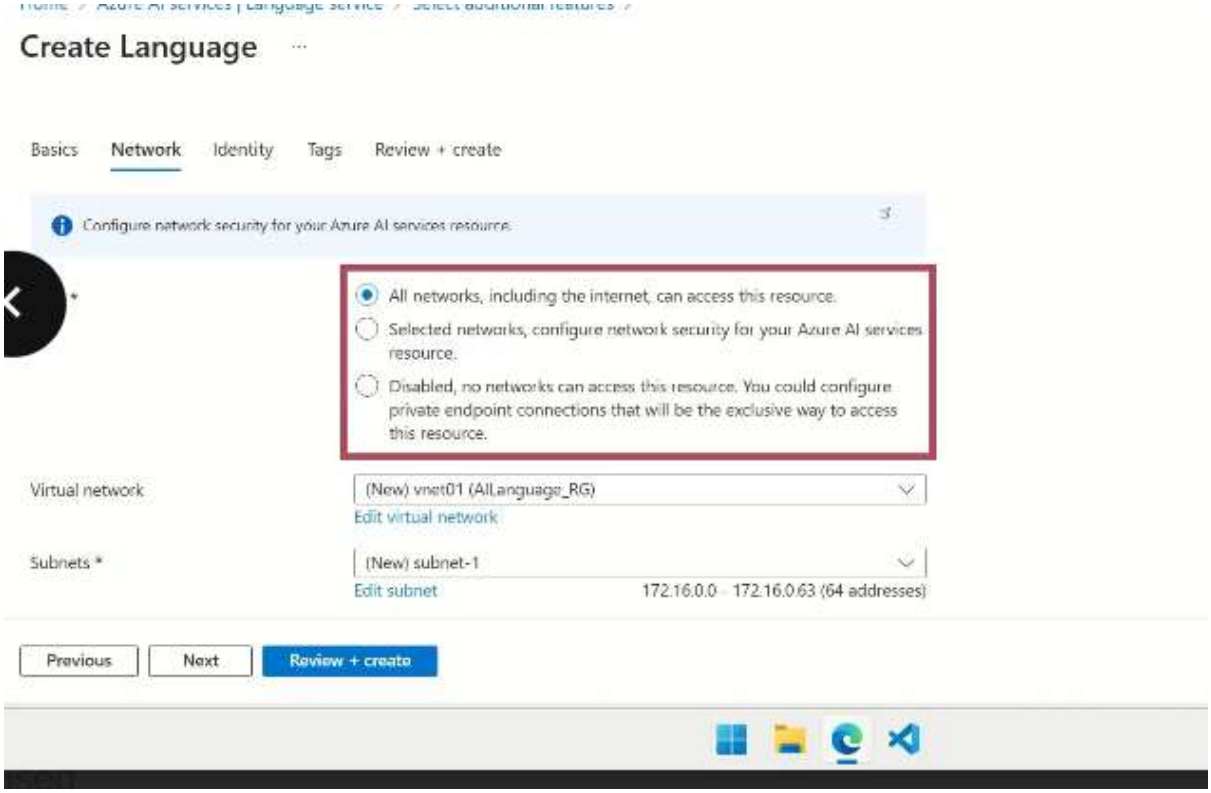
- ➔ Extract key details to identify the core issue and categorize the inquiry appropriately
- ➔ Deploy an Azure AI Language resource
- ➔ Extract key phrases
- ➔ Extract custom named entities
- ➔ Implement entity linking
- ➔ Extract personal identifiable information

➔ Azure Al Language Resource

- Key Phrase Extraction: Automatically identifies the main concepts or topics within text.
- Named Entity Recognition : Identifies and classifies real-world objects like people, locations, and organizations in unstructured data.
- Entity Linking
- PII Extraction : Detects and redacts Personally Identifiable Information to ensure data
- Summarization
- Language Detection

➔ Create language by going to azure ai studio

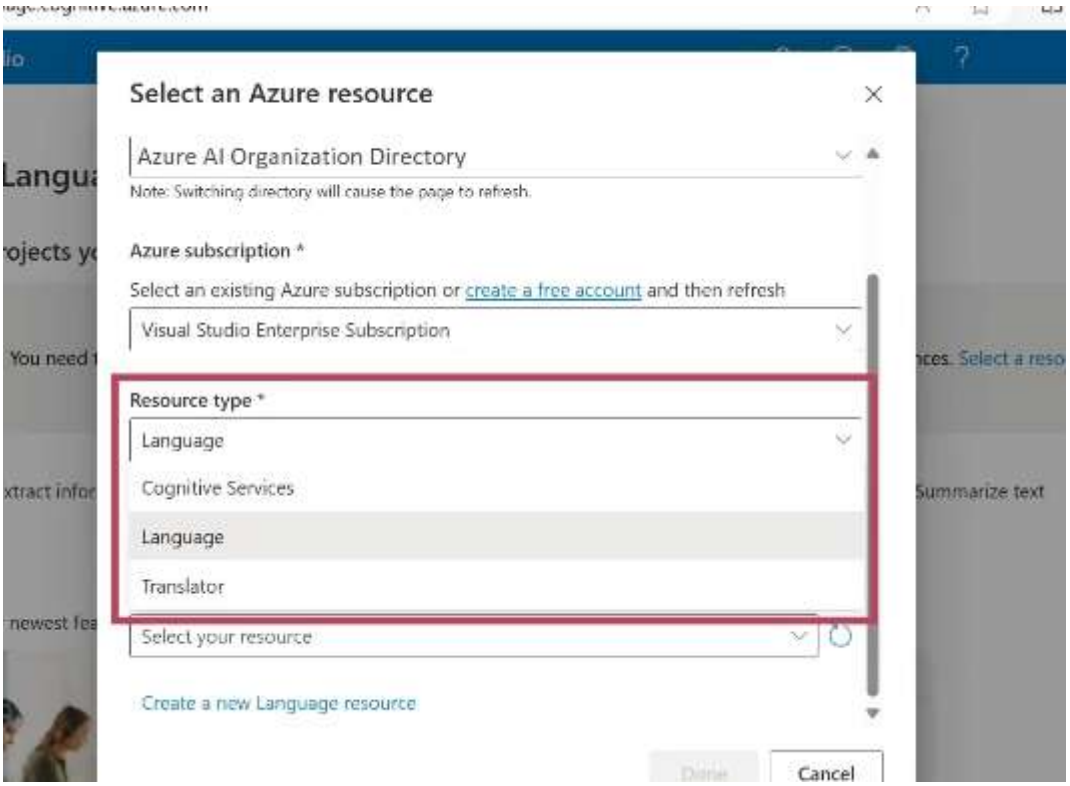




➔

➔ After it is deployed you can use it

➔ Open Lnagugae studio on Azure AI



➔

➔ Use tool -> Extract key phrases

➔ Then you can upload and extract

➔ Then writing code on vscode

➔ First is Text Analysis

- Import the required modules

```
#   pip install python-dotenv

import sys
import os
import json
from dotenv import load_dotenv
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential
```

- Creating a functon endpoint for retrive key and service endpoint from .env and for linking azure ai authentication
- The code to extract key phrase from the set of documnet

```
def extract_key_phrases(client, documents):
    try:
        response = client.extract_key_phrases(documents=documents)
        result = [doc for doc in response if not doc.is_error]
        return result
    except Exception as err:
        print(f"Encountered exception. {err}")
        return []
```

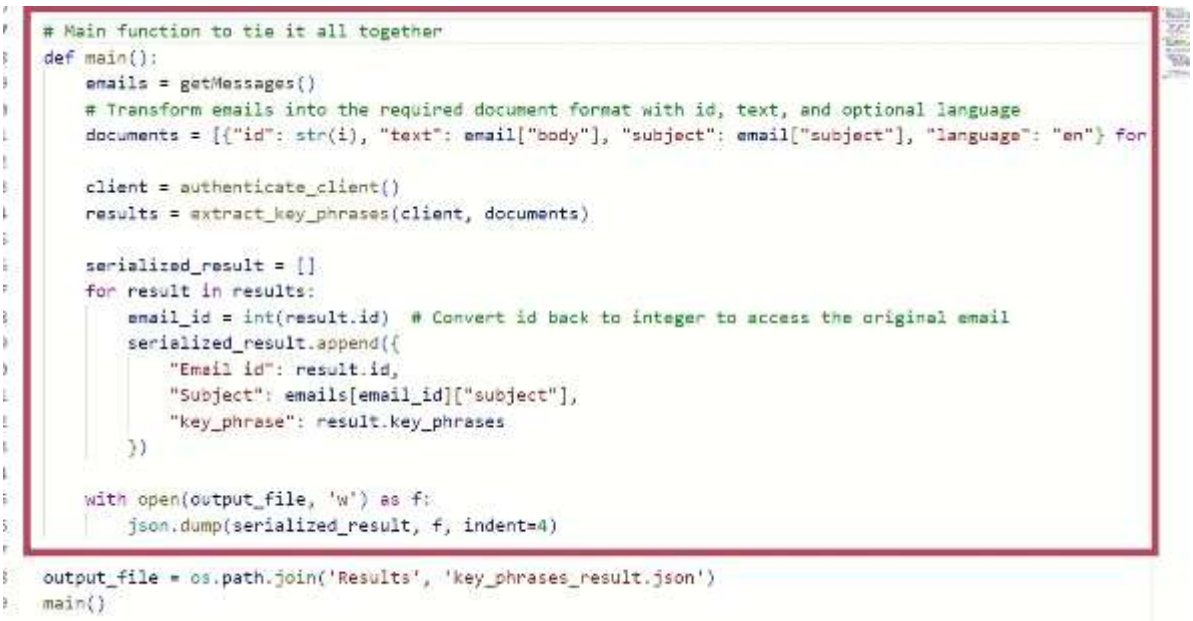
Function

Purpose: The extract_key_phrases function is designed to take a client

object and a list of documents, then process them to find significant phrases [1].

- **Key Phrase Extraction**: The highlighted line (response = client.extract_key_phrases(...)) actually calls the service to perform the analysis on the provided documents [1].
- **Error Handling**: The try...except block is included to catch any exceptions that might occur during the API call, printing the error and returning an empty list if a failure occurs [1].
- **Result Filtering**: The result = [...] line filters the API response to include only successfully processed documents, excluding those that resulted in an error [1].

- Python script defines a main() function that orchestrates an email processing workflow, likely using an Azure Cognitive Services SDK for key phrase extraction.



```
# Main function to tie it all together
def main():
    emails = getMessages()
    # Transform emails into the required document format with id, text, and optional language
    documents = [{"id": str(i), "text": email["body"], "subject": email["subject"], "language": "en"} for

    client = authenticate_client()
    results = extract_key_phrases(client, documents)

    serialized_result = []
    for result in results:
        email_id = int(result.id)  # Convert id back to integer to access the original email
        serialized_result.append({
            "Email id": result.id,
            "Subject": emails[email_id]["subject"],
            "key_phrase": result.key_phrases
        })

    with open(output_file, 'w') as f:
        json.dump(serialized_result, f, indent=4)

output_file = os.path.join('Results', 'key_phrases_result.json')
main()
```

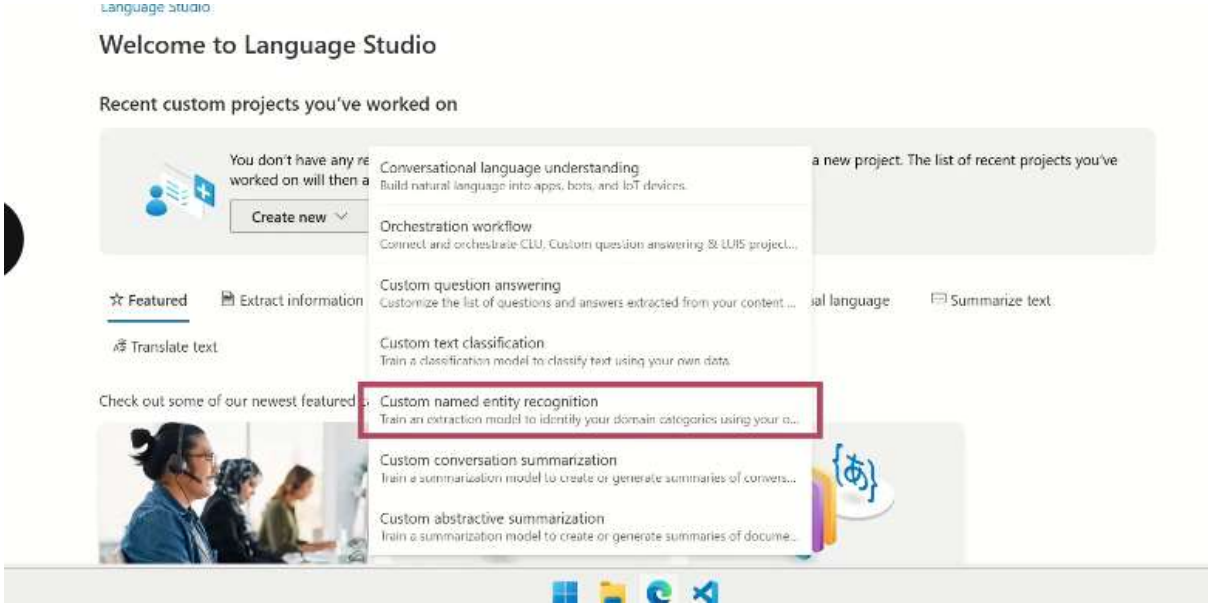
- **Data Retrieval:** It calls getMessages() to fetch a list of email objects.
- **Data Transformation:** It converts those emails into a specific documents format (a list of dictionaries containing id, text, subject, and language) required by the API.
- **Processing:**
  - Authenticates a client.
  - Sends the documents to extract_key_phrases() to analyze the text.
- **Serialization:**
  - Iterates through the results.

- Maps the analysis back to the original email data using the id.

- Constructs a new list of dictionaries containing the Email ID, Subject, and the extracted Key Phrases.
- **Output:** Saves the final structured data into a JSON file located at Results/key_phrases_result.json

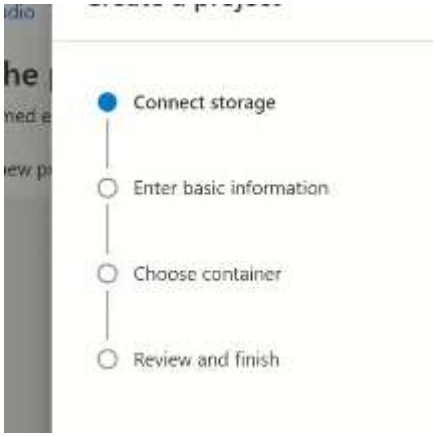
➔ Language service provide entity recognizition , we can navigate to named entity recognition

➔ Return back to language studio and create a new



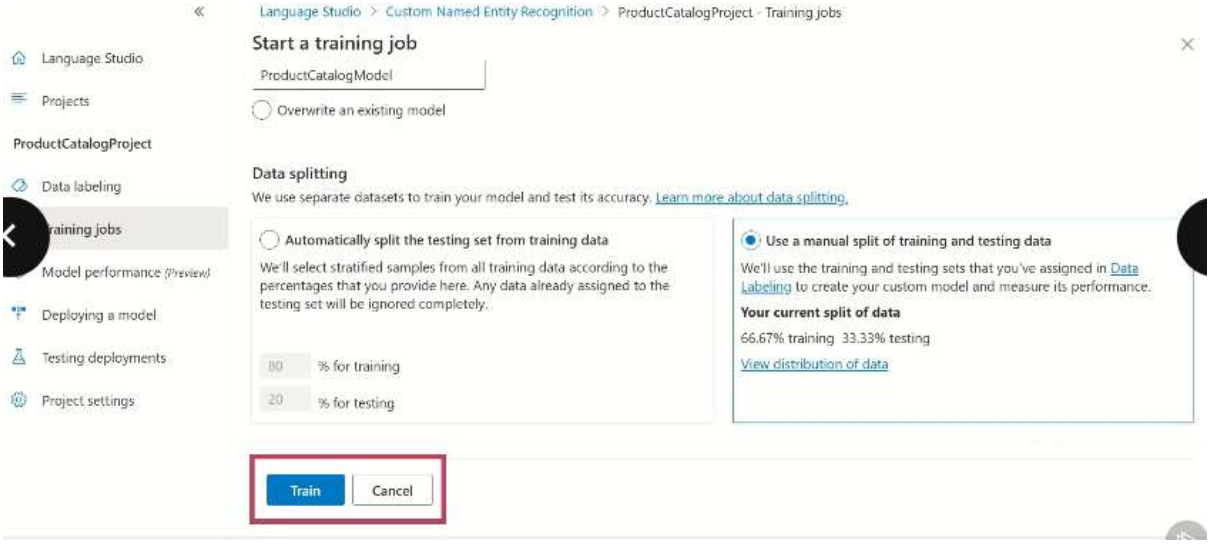
- **Conversation Understanding:** Features conversational language understanding (CLU) to help build interactive bots and natural language applications.
- **Question Answering:** Includes tools for customizing how bots extract answers from content.
- **Text Analysis:** Allows for custom named entity recognition and classification to train models to identify domain-specific categories. [1, 2, 3]

➔ It is a four stage process :



➔ Labelling you input under some label categories and dividing them for extracting testing training data differently

➔ Training the job



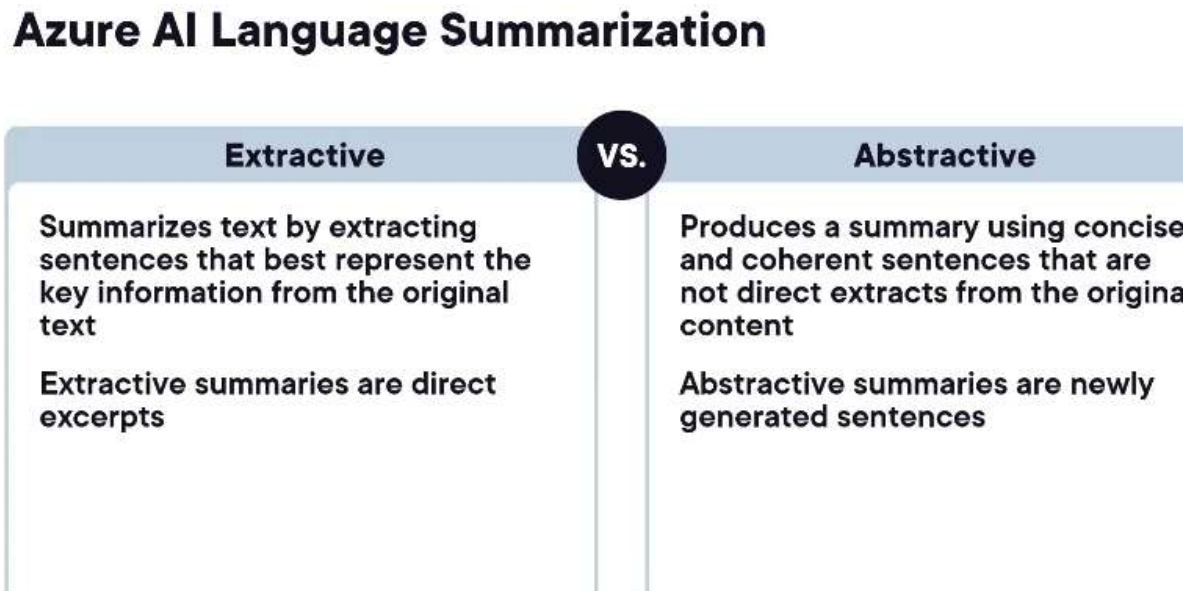
➔ Adding the product created

```
29  def extract_custom_named_entities(client, documents):
30      try:
31          operation = client.begin_recognize_custom_entities(
32              documents,
33              project_name="ProductCatalogProject",
34              deployment_name="ProductCatalogDeployment"
35          )
```

Demo: Help to understand customers issues by summarizing long emails

➔ Implementing and extracting summary from email

➔ This can be done in two ways:



➔ Open language studio again and navigate to summarization information tryout

➔ And you can summarize the text

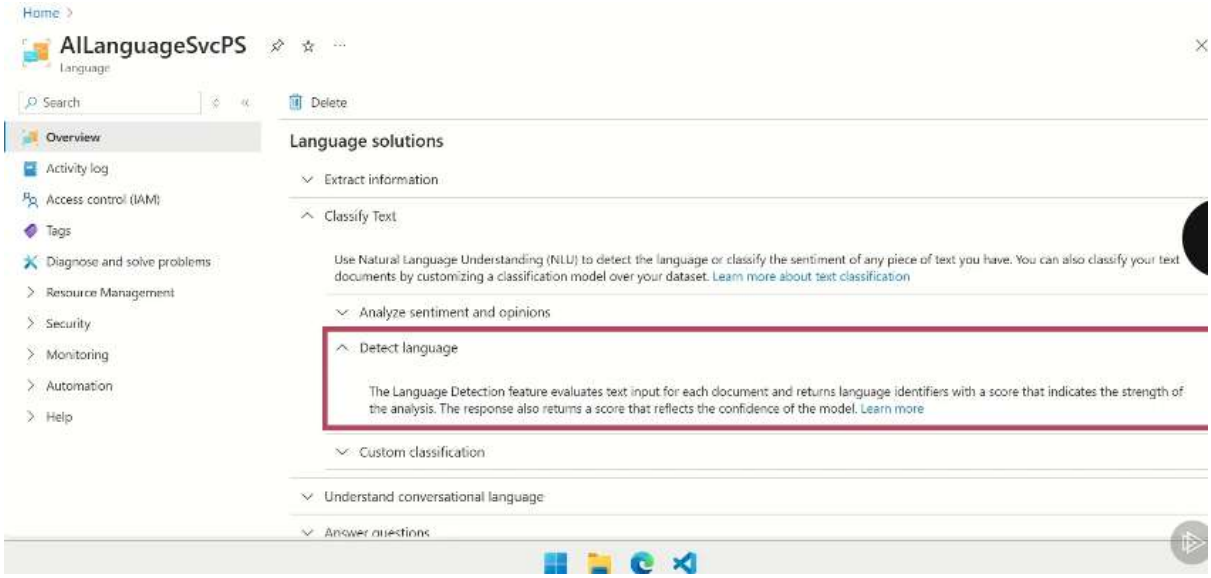
- Then include this features within project

```
2 - Summarization > summarization.py > main
27
28  def summarization(client, documents):
29      try:
30          poller = client.begin_abstract_summary(documents)
31          result = poller.result()
32          docs = [doc for doc in result if not doc.is_error]
33
34          results = []
35          for i, result in enumerate(docs, start=1):  # Start indexing from 1
36              if result.kind == "AbstractiveSummarization":
37                  summary = " ".join([summary.text for summary in result.summaries])
38                  results.append({
39                      "Email id": i,
40                      "Subject": next(d['subject'] for d in documents if d['id'] == result.id),
41                      "Summary": summary
42                  })
43              elif result.is_error:
44                  print(f"Error: {result.error.code} - {result.error.message}")
```

- **Asynchronous Processing:** The code uses client.begin_abstract_summary(documents) to initiate an asynchronous operation, allowing the application to continue running while the service processes the documents. [1]
- **Result Retrieval:** poller.result() blocks execution until the summarization task is complete and retrieves the results. [1]
- **Data Structuring:** The code iterates through the successfully processed documents, extracts the generated summaries, and maps them back to the original document subjects using a dictionary format. [1]

- **Error Handling:** A basic error-checking mechanism (if result.is_error) is included to catch and print error codes and messages if the summarization fails for a specific document. [1]

Implementing text language detection capabilities:



Using using azure Ai language studio and there using can define the language that it has to be detecting in and passing the language and deciding the output

This detection of language can be used in email summarization and is used for detecting the language used in email and extracting the language content.